Longest Subarray With Sum

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The problem

- Non-negative ints; [start, end] of the longest subarray summing to target, or [] if none.
- Sliding window.

Intuition

- Values are non-negative, so a sliding window grows to add and shrinks to subtract.

Approach

- 1. Expand the right edge adding values
- 2. Shrink from the left while the sum is too big
- 3. Record the longest window equal to target

Java solution

```java
public class LongestSubarrayWithSum {
    public static int[] longestSubarrayWithSum(int[] a, int target) {
        int[] best = {};
        int l = 0, sum = 0;
        for (int r = 0; r < a.length; r++) {
            sum += a[r];
            while (sum > target && l < r) sum -= a[l++];
            if (sum == target && (best.length == 0 || r - l > best[1] - best[0])) best = new int[]{l, r};
        }
        return best;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Values are non-negative, so a sliding window grows to add and shrinks to subtract.
- Complexity: O(n)